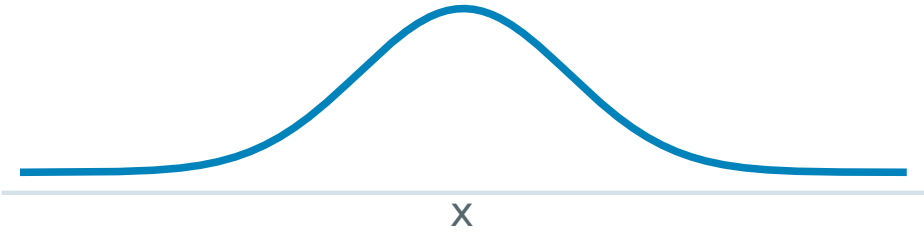
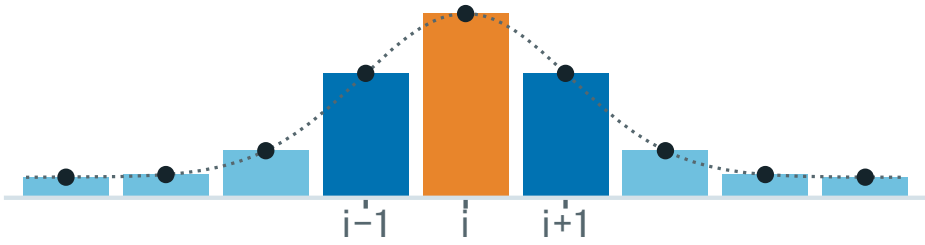
1 Continuous field $u(t_k, x)$



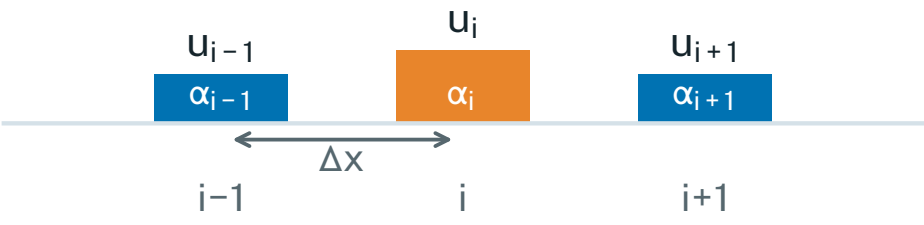
2 Discretise: grid values u_i^k



3 Classical stencil at cell i

$$f_i = \frac{1}{\Delta x^2} \sum_j \alpha_j u_j$$

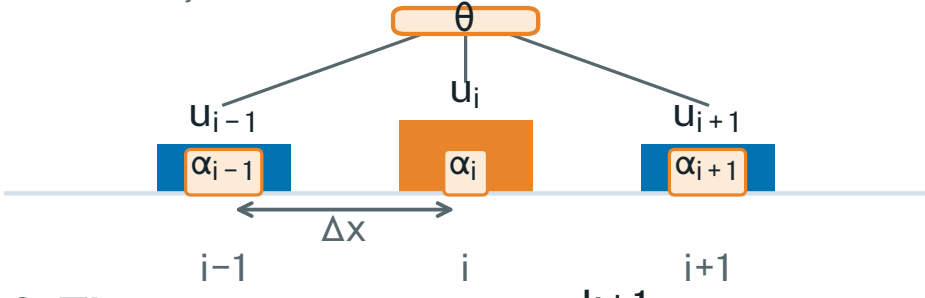
weights α_j fixed



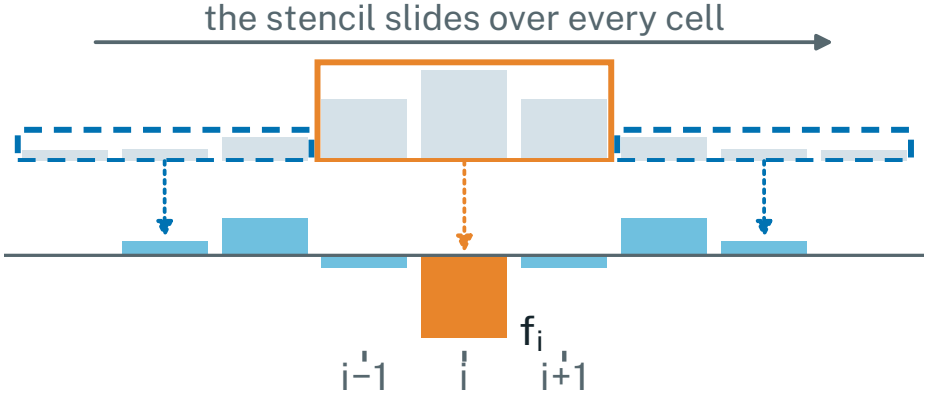
4 Learned stencil at cell i

$$f_i = \frac{1}{\Delta x^2} \sum_j \alpha_j(u;\, \theta)\, u_j$$

weights α_j computed from the values by a network θ

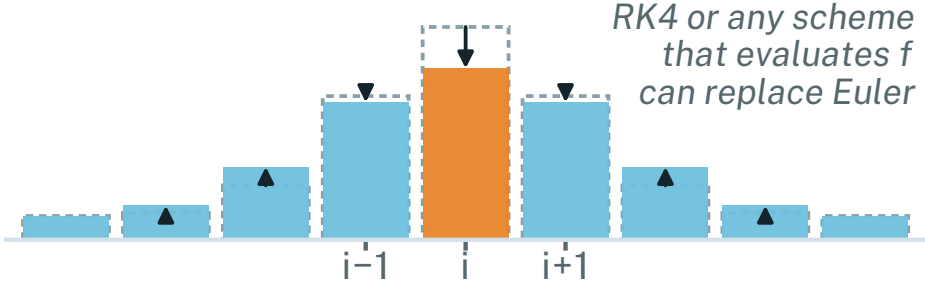


5 Slide the stencil: rates f_i



6 Time step: next state u_i^{k+1}

$$u_i^{k+1} = u_i^k + \Delta t\, f_i$$



RK4 or any scheme that evaluates f can replace Euler